Rocky Shao

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EDUCATION

The Ohio State University ***Expected Graduation: May 2028***
B.S. in Computer Engineering **GPA:** 4.0

WORK EXPERIENCE

Undergraduate Research Assistant (Power Electronics) ***May 2026 – Present***
Center for High Performance Power Electronics (Prof. Jin Wang), The Ohio State University, Columbus, OH

- Collaborate with 2 PhD and 3 Master's researchers from Ohio State and Miami University on a GE Aerospace and Ohio Federal Research Network (OFRN) sponsored project developing a smart connector for gas-insulated high-voltage DC cables with integrated partial discharge (PD) detection for electrified aircraft using an onboard physics-informed neural network (PINN).
- Architect the project's high-performance, resource-efficient digital signal processing (DSP) pipeline and inverter controller.
- Presented the paper "Series Connection of GaN-based Monolithic Bidirectional Switches for 800-V DC Applications" at the IEEE Energy Conversion Congress and Exposition (ECCE) in October 2026.

Research Project: Discrete Fourier Transform (DFT) Hardware Accelerator ***July 2026 – Present***

- Design, simulate, and synthesize a custom DSP accelerator in Verilog and SystemVerilog on an Altera Cyclone IV FPGA using ModelSim and Quartus Prime, ingesting a 100 MHz data stream and achieving 0.04% error compared to MATLAB.
- Evolve the architecture from a targeted multi-bin parallel Discrete Fourier Transform (DFT) to an SDRAM-backed serial replay controller for arbitrary harmonics, optimizing FPGA compute resource requirements from linear growth to constant usage.
- Replace a non-scalable embeded RAM (BRAM) twiddle factor table with a 128-entry coarse Lookup Table (LUT) paired with an on-the-fly Q1.15 fixed-point linear interpolation to minimize FPGA memory utilization.
- Architect an external SDRAM interface to buffer continuous ADC sample streams without exhausting on-chip memory.

Research Project: PWM Inverter Controller (TI C2000 & Cyclone IV FPGA) ***June 2026 – Present***

- Design dual-chip embedded firmware across a TI C2000 MCU and an Altera Cyclone IV FPGA to drive a 4-channel power inverter.
- Engineer a custom 32-bit SPI protocol with 2-FF Clock Domain Crossing (CDC) synchronizers and edge detection on the FPGA, safely transferring frequency, duty cycle, duration, and deadband registers.
- Implement an FPGA PWM generator with programmable deadband and zero-state fail-safe shutdown logic, eliminating shoot-through risk across complementary power switch pairs.
- Program a non-blocking UART command-line interface on the MCU for real-time calibration; verify sub-microsecond deadtime accuracy on high-bandwidth oscilloscopes.

EXTRACURRICULAR ACTIVITIES

Autonomous Vehicle Software Team Member (Buckeye AutoDrive) ***Sep 2025 – July 2026***

- Selected as 1 of 3 first-year students on Ohio State's 150+ member, primarily graduate, SAE Level 4 autonomous vehicle team for the AutoDrive Challenge II.
- Engineered a bi-directional Robot Operating System (ROS 2) to Transmission Control Protocol (TCP) middleware bridge between the central Intel compute server and an NVIDIA Jetson touchscreen Human-Machine Interface (HMI) to ensure a reliable autonomous engagement sequence and intuitive vehicle control.
- Refactored monolithic telemetry code into decoupled, asynchronous ROS 2 nodes, multiplexing GPS, waypoint, and perception streams to allow functionality scaling while isolating safety-critical engage commands.
- Resolved a critical vehicle Finite-State Machine crash by engineering a state-aware single-pulse dispatch filter that eliminated multiple attempts to enter the engage state from duplicate engage commands during autonomous mode transitions.
- Eliminated live-competition HMI screen freezes caused by 60 Hz perception bursts through frame downsampling and unknown-object filtering.
- Configured systemd services and automated ROS 2 launch pipelines for reliable vehicle boot and fault recovery.
- Presented Concept Design Event in competition, achieving score 66/75.

TECHNICAL SKILLS

- **Programming & HDLs:** Python, C, C++, MATLAB, Verilog, SystemVerilog
- **Hardware & Protocols:** FPGA (Altera Cyclone IV), MCU (TI C2000), SPI, UART, TCP/IP, Protobuf
- **Tools & Software:** ModelSim, Quartus Prime, ROS 2, Git, Linux, Digital Oscilloscopes
- **Core Competencies:** Digital Logic Design, DSP, Hardware/Software Co-Design, Real-Time Firmware, RTL Verification